

Python for Data Analysis – Day 1: Introduction to Python

WHAT IS PYTHON?

Python is a high-level, interpreted, general-purpose programming language.

It was created by Guido van Rossum and released in 1991.

Python is one of the most popular programming languages because it is:

- Easy to learn
 - Easy to read
 - Powerful
 - Free and Open Source
 - Used in Data Analysis, AI, Machine Learning, Automation, Web Development, and Cyber Security
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Why Should Data Analysts Learn Python?

Imagine you have:

Task	Without Python	With Python
Clean 1 lakh rows	2 Hours	2 Minutes
Merge 10 Excel Files	1 Hour	10 Seconds
Create Reports	Manual	Automatic
Dashboard Data Preparation	Difficult	Easy

Python helps automate repetitive tasks and analyze large datasets efficiently.

Real-Life Uses of Python

1. DATA ANALYSIS

Analyze sales, customer, and employee data.

Example:

```
sales = [10000, 15000, 20000]

print(sum(sales))
```

Output:

```
45000
```

2. ARTIFICIAL INTELLIGENCE (AI)

Used in:

- ChatGPT
- Face Recognition
- Voice Assistants
- Recommendation Systems

3. MACHINE LEARNING

Used for:

- Sales Prediction
- Customer Churn Prediction
- Fraud Detection

4. AUTOMATION

Automate:

- Excel Reports
- Emails
- File Management
- Data Entry

5. BUSINESS INTELLIGENCE

Works with:

- Excel
 - SQL
 - Power BI
 - Tableau
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Career Opportunities After Learning Python

DATA ANALYST

Skills:

- Excel
- SQL
- Python
- Power BI

Typical work:

- Reporting
 - KPI Analysis
 - Dashboard Development
-

BUSINESS ANALYST

Skills:

- Excel
- SQL
- Python

Typical work:

- Business Reports
 - Requirement Analysis
 - Data Insights
-

MIS EXECUTIVE

Skills:

- Excel
- VBA
- Python

Typical work:

- Automated Reports
 - Data Management
-

DATA SCIENTIST

Skills:

- Python
- Statistics
- Machine Learning

Typical work:

- Prediction Models
 - AI Systems
-

DATA ENGINEER

Skills:

- Python
- SQL
- ETL Tools

Typical work:

- Data Pipelines
 - Data Warehousing
-

Python Installation

STEP 1

Download Python from:

[Python Official Website](#)

STEP 2

Install Python.

✓ Tick:

Add Python to **PATH**

STEP 3

Verify Installation

Open Command Prompt:

```
python --version
```

Output:

```
Python 3.x.x
```

Python IDEs

JUPYTER NOTEBOOK

Best for:

- Data Analysis
 - Machine Learning
-

VS CODE

Best for:

- Professional Development
 - Python Projects
-

GOOGLE COLAB

Best for:

- Learning Online
 - AI Projects
-

First Python Program

```
print("Hello World")
```

Output:

```
Hello World
```

EXPLANATION

`print()` displays output on screen.

Comments

Comments are ignored by Python.

SINGLE-LINE COMMENT

```
# This is a comment
```

```
print("Hello")
```

MULTI-LINE COMMENT

```
"""
```

```
This is  
a multi-line  
comment
```

```
"""
```

Variables

Variables store data.

Example:

```
name = "Umesh"
```

```
age = 25
```

```
print(name)
```

```
print(age)
```

Output:

```
Umesh
```

```
25
```

Variable Naming Rules

CORRECT

```
name
```

```
salary
```

```
employee_name
```

```
age1
```

INCORRECT

```
1name
```

```
employee-name
```

```
for  
class
```

Data Types

STRING (STR)

Stores text.

```
name = "Umesh"  
  
print(type(name))
```

Output:

```
<class 'str'>
```

INTEGER (INT)

Stores whole numbers.

```
age = 25  
  
print(type(age))
```

Output:

```
<class 'int'>
```

FLOAT

Stores decimal values.

```
salary = 25000.50  
  
print(type(salary))
```

Output:

```
<class 'float'>
```

BOOLEAN

Stores True or False.

```
is_active = True

print(type(is_active))
```

Output:

```
<class 'bool'>
```

Taking Input

```
name = input("Enter Name: ")

print(name)
```

Output:

```
Enter Name: Umesh
Umesh
```

Numeric Input

```
age = int(input("Enter Age: "))

print(age)
```

Formatted Output

METHOD 1

```
name = "Umesh"

print("Welcome", name)
```

Output:

```
Welcome Umesh
```

METHOD 2 (F-STRING)

```
name = "Umesh"
age = 25
```



```
print(f"My Name is {name} and Age is {age}")
```

Output:

```
My Name is Umesh and Age is 25
```

Type Conversion

STRING TO INTEGER

```
age = "25"
```

```
age = int(age)
```

```
print(age)
```

INTEGER TO STRING

```
salary = 25000
```

```
salary = str(salary)
```

```
print(salary)
```

Real Business Example

Employee Information

```
employee_name = "Rahul"
```

```
department = "Sales"
```

```
salary = 45000
```

```
print(employee_name)
```

```
print(department)
```

```
print(salary)
```

Output:

```
Rahul
```

```
Sales
```

```
45000
```

Mini Project 1: Employee Card

```
name = input("Enter Employee Name: ")
emp_id = input("Enter Employee ID: ")
salary = float(input("Enter Salary: "))

print("\nEmployee Details")
print("Name:", name)
print("ID:", emp_id)
print("Salary:", salary)
```

Mini Project 2: Age Calculator

```
birth_year = int(input("Enter Birth Year: "))

age = 2026 - birth_year

print("Age =", age)
```

Practice Questions

BASIC

1. Print your name.
2. Print your city.
3. Print your age.
4. Print today's date.
5. Print "Welcome to Python".

VARIABLES

6. Create a variable called name.
7. Create a variable called city.
8. Create a variable called salary.
9. Print all variables.
10. Create five variables and display them.

DATA TYPES

11. Create a string variable.
12. Create an integer variable.
13. Create a float variable.

14. Create a boolean variable.
15. Check datatype using `type()`.

INPUT

16. Take employee name as input.
17. Take employee ID as input.
18. Take salary as input.
19. Take city as input.
20. Display all details.

TYPE CONVERSION

21. Convert "100" into integer.
 22. Convert 500 into string.
 23. Convert 25.5 into integer.
 24. Convert 10 into float.
 25. Check datatype before and after conversion.
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Assignment (Data Analyst Level)

Create a program that takes:

- Employee Name
- Employee ID
- Department
- Salary

and displays:

EMPLOYEE SUMMARY

```
Employee Name : Rahul
Employee ID   : E101
Department    : Sales
Salary        : 45000
```

INTERVIEW QUESTIONS

1. What is Python?
2. Who created Python?
3. What are the advantages of Python?
4. What is a variable?
5. What are data types?

6. Difference between int and float?
7. What is type conversion?
8. What is input()?
9. What is print()?
10. Why is Python important for Data Analysts?